

# Joshua Kennedy

Cloud Platform Engineer  
Resides in Scotland

## Experience

### Cloud Platform Engineer

- NatWest - October 2023 - Present
- > Engineered strategic on-premise Container-as-a-Service (CaaS) capabilities supporting 32 OpenShift clusters across four environments for approximately 300 engineering teams.
  - > Developed platform engineering capabilities through policy-driven automation and GitLab CI/CD, delivering centralised GitOps orchestration, HashiCorp Vault integration and customer guardrails.
  - > Designed reusable platform capabilities consumed by ~1200 applications, reducing operational complexity while improving consistency and self-service.
  - > Collaborated within a 20-person platform engineering team alongside principal engineers, architects and stakeholders to deliver shared platform capabilities in a regulated financial services environment.
  - > Continuously evaluated, prototyped and implemented improvements to platform tooling, automation and operational workflows to increase platform maturity and developer productivity.

### Cloud Automation Engineer

- NatWest - September 2021 - October 2023
- > Joined a small R&D team focused on expanding the bank's on-premise cloud computing capabilities.
  - > Built experience across OpenShift, Kubernetes, container orchestration, Linux and cloud platform automation.
  - > Delivered proof-of-concept work for certificate automation using cert-manager and centralised secret management using HashiCorp Vault.
  - > Collaborated across teams to turn emerging platform requirements into working prototypes and executable patterns.
  - > Supported automation and CI/CD-aligned delivery practices to reduce manual effort and increase consistency across platform operations.

### Middleware Systems Engineer

- NatWest - February 2019 - September 2021
- > Built automation to streamline common support tasks for deployment orchestration tooling within a feature team.
  - > Automated agent upgrade workflows, enabling over 1,600 agents to be upgraded across approximately 50 teams within six months.
  - > Supported users and administrators of deployment orchestration platforms, producing documentation and running workshops to upskill a wider support team.
  - > Improved platform support processes by combining scripting, documentation, incident analysis and close collaboration with consuming teams.

### Middleware Technical Analyst

- RBS - January 2018 - February 2019
- > Delivered technical support for Java-based enterprise middleware, resolving over 150 individual incidents.
  - > Developed infrastructure support scripts and helped stand up deployment orchestration tooling.
  - > Built practical experience with REST API integration, high-availability clustering and production support in a regulated banking environment.

### WebSphere Technical Apprentice

- RBS - January 2016 - January 2018
- > Developed KornShell automation to replace a manual licensing process across approximately 5,000 servers.
  - > Audited and improved certificate renewal tracking workflows, building skills in Visual Basic, SQL and enterprise middleware administration.
  - > Completed Modern and Graduate Apprenticeship pathways while working in production financial services technology teams.

## Personal Projects

### tapo-onboarding

Shell / Reverse Engineering / IoT  
Reverse engineered TP-Link's encrypted provisioning protocol to create one of the first open-source toolchains capable of onboarding modern Tapo cameras without vendor cloud services, solving a problem previously lacking a practical community solution.

### Stateful

JavaScript / Pebble / REST APIs  
Developed a generic REST client for Pebble that enabled arbitrary REST APIs to be modelled and controlled from the watch without assumptions about the underlying service, addressing a capability absent from existing applications on the platform.

### restate-go

Go / Home Automation / APIs  
Developing a local-first home automation platform that prioritises ownership, privacy and reliability over cloud-dependent architectures, exploring an alternative to conventional commercial ecosystems.

### Snowdon II WiFi

C++ / Raspberry Pi Pico / REST APIs / Infrared  
Engineered an embedded hardware modification that adds Wi-Fi and a REST API to a consumer soundbar through infrared protocol emulation and state inference, creating capabilities unavailable from the manufacturer.

## Education and Certification

### Certificate of Higher Education in Engineering

April 2023 - October 2025

### Red Hat OpenShift Administration II: Configuring a Production Cluster (DO280)

October 2024

### Red Hat Certified Specialist in Containers and Kubernetes

July 2022

### Graduate Apprenticeship Scotland in IT & Telecommunications Professionals

January 2017 - November 2017

### Modern Apprenticeship Scotland in IT & Telecommunications Professionals

January 2016 - November 2016

### HND Computer Games Development

August 2013 - August 2015

## Technical Skills

Platform Engineering

Software Engineering

OpenShift / KubernetesAutomation

CI/CDLinuxInfrastructure as Code

GoShell / Bash / KornShellREST APIs

PythonJavaScriptEmbedded C / C++

Security & Policy Controls

## Professional Skills

Technical Leadership

Stakeholder Communication

Cross-functional Collaboration

Problem SolvingContinuous Improvement

Mentoring

## Contact

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